

Knowledge Representation and Reasoning

POLITEHNICA Bucharest
Department of Computer Science
Fall 2026

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Lecture 1

Lecture outline

- Course goals
- Grading
- Textbooks and readings
- Syllabus
- What is KR and what is Reasoning?
- KR&R Challenges
- Models of KR

Course goals

- Provide an overview of the main representation models developed within AI, their key concepts and the associated inference & learning methods
- Acquiring skills in representing knowledge
- Being able to read and understand research literature in the area of KR&R
- Being able to complete projects in this research area

Grading

- ***Course grades***

Mid-term exam	20%
Final exam	30%
Homeworks	25%
Laboratory	25%
Course tests	10% - Bonus points

- ***Requirements:*** min 7 lab attendances, min 50% of term activity (mid-term ex, homeworks, lab)
- Solving lab work must be presented mandatory in the next week.
- ***Academic Honesty Policy***
  It will be considered an honor code violation to give or use someone else's code or written answers, either for the assignments or exam tests, or to use LLMs without explicitly indicated which parts were produced by those models.

Textbooks and Readings

- ***Textbooks***
  - *Artificial Intelligence: A Modern Approach (4rd Edition)*
    by Stuart Russell and Peter Norvig Prentice Hall, 2021 – Part IV and V
    http://aima.cs.berkeley.edu/global-index.html
  - *Probabilistic Graphical Models. Principles and Techniques* by Daphne Koller and Nir Friedman, MIT Press, 2009 – Chapters 1,2,3,9,10 & 19
    https://mitpress.mit.edu/books/probabilistic-graphical-models
  - *Deep Learning*
    by Ian Goodfellow and Yoshua Bengio and Aaron Courville, MIT Press, 2016
    https://www.deeplearningbook.org/
  - *Foundations of Large Language Models*
    by Tong Xiao and Jingbo Zhu, an open-access work on arXiv, 2025
    https://arxiv.org/pdf/2501.09223
- ***Readings***
  - Reading materials will be assigned to you.
  - You are expected to do the readings before the class

Syllabus

1. **Knowledge representation models - overview**
2. **Symbolic representations**
   - Ontologies & OWL
   - Knowledge graphs
   - Rule based representations
3. **Probabilistic graphical models**
   - Bayesian networks
   - Hidden Markov models
4. **Nonsysmbolic representation models**
   - LLMs representations
   - SSL representations & learning
   - Transfer learning
5. **Explainable AI for black box models**

Lecture 1

Introduction

- Knowledge Representation Models

1. What is KR?

What is knowledge and KR?

- We understand by **"knowledge"** all kinds of facts about the world.
- Intuitively, **knowledge representation** is the way an AI system stores, organizes, and structures knowledge about the world so it can use that knowledge to reason, make decisions, and solve problems
- With different **models of KR** the AI system interprets the data and can reason about it, and is thus able to develop **intelligent artifacts** (based on software, hardware or both).

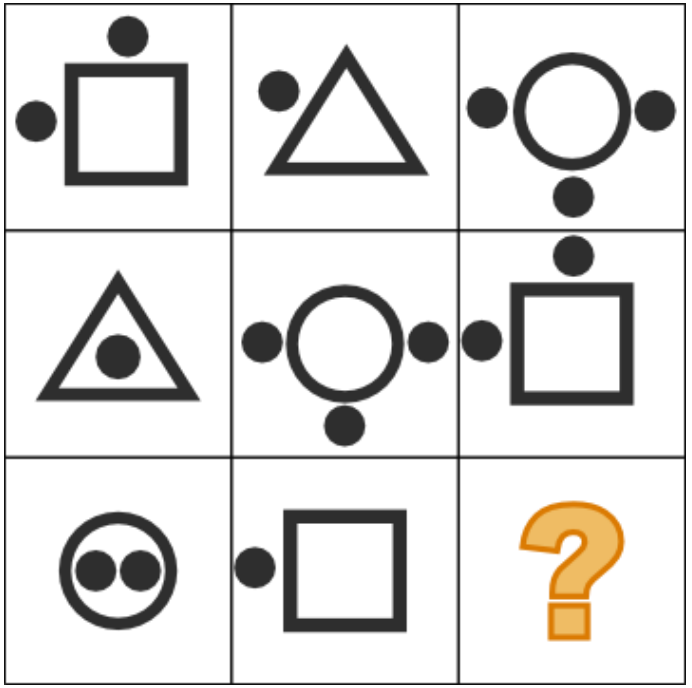
A definition of KR

Randall Davis, Howard Shrobe, Peter Szolovits, MIT

- A **knowledge representation** is most fundamentally **a surrogate**, a *substitute* for the thing itself, used to enable an entity to determine new facts or arrive at a decision by reasoning about the world.
- It is **a set of ontological commitments**, i.e., an answer to the question: *In what terms should I think about the world?*

2. What is Reasoning?

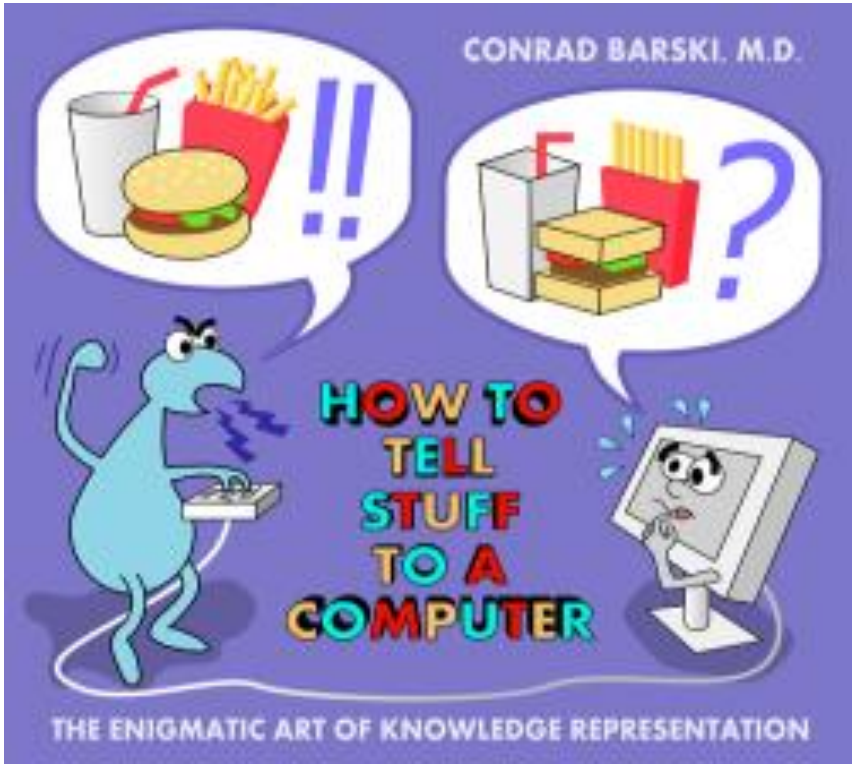
- **In AI, reasoning** is the use of knowledge representations in order **make inferences, draw conclusions, and solve problems** based on available knowledge, rules, or evidence



- **Inference** – a form of reasoning
  - **General** reasoning/ inference models
  - **Specific** inference models – specific to each model of representation
  - **Symbolic** inference
  - **Non-symbolic** inference

3. KR&R Challenges

- Challenges of KR&R:
  - the ability of a system to achieve **computational efficiency** for **different types of inferences & learning**
  - the ability to represent and manipulate **uncertain knowledge** and information
  - capacity to **explain** how it arrived to a **decision**



4. Models of KRR

Symbolic models

- Logical models
  - First order logic
  - Higher order logics
  - Description logics
- Structured models
  - Semantic networks
  - Knowledge graphs
  - Ontologies
- Rule based models
  - Rule based systems (OPS5, CLIPS, DROOLS)
  - Sematic rule systems (RuleML, SWRL)
- Decision models (symbolic)

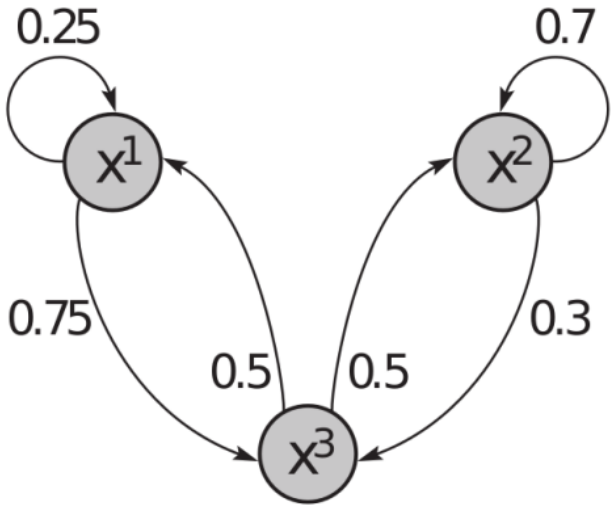


"It may be a model, Captain, but it's highly illogical."

Models of KRR

Probabilistic models

- Bayesian networks
- Markov chains
- Markov Random Fileds
- Hidden Markov Models
- Markov Decision Process
- Partially Observable Markov Decision Process



Models of KRR

Non-symbolic models

- Regression models
- Clustering models
- Fuzzy logic and systems: stands in the middle between s and non-s
- Genetic models
- Distributed representations with “embeddings”
- Deep Neural networks
- Large Language Models (LLMs)
- Vision-Language Models (VLMs)

Symbolic models

Logical models

- Use a **symbolic notation** to express facts and logical sentences about the world in a formal way
- Logic-based representations use formal logic to express knowledge in terms of propositions, predicates, and logical relationships between them

- Logical models
  - First order logic
  - Higher order logics
  - Description logics

Logical models

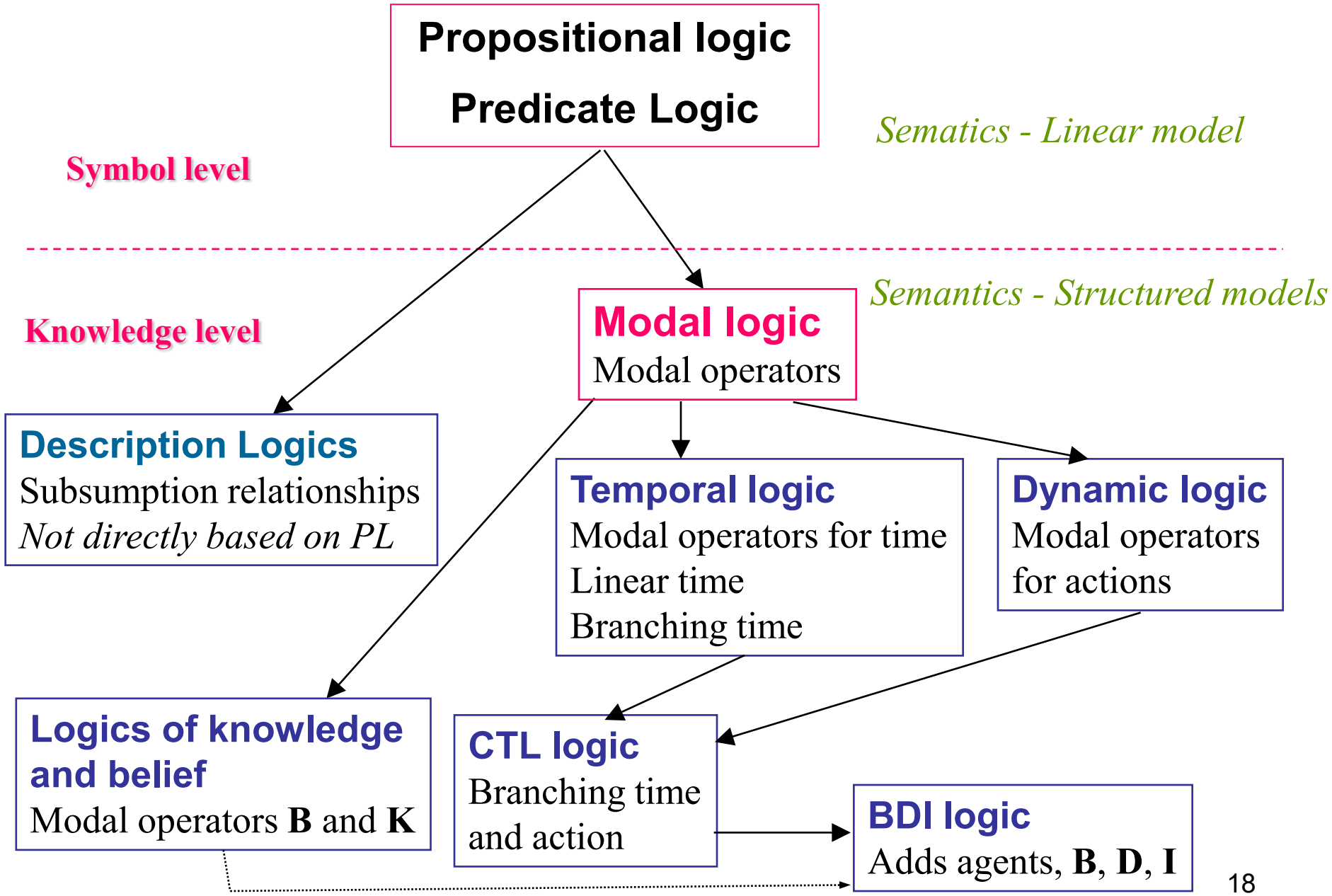
knowledge	propositional	first-order
Paul is a man	a	man(Paul)
Bill is a man	b	man(Bill)
men are mortal	c	$(\forall x)$ (man(x) $\supset$ mortal(x))

First order logic

knowledge	first-order	second-order
smaller is transitive	$(\forall x)((\forall y)((\forall z)((<(x,y) \wedge <(y,z) \supset <(x,z)))))$	transitive(<)
part-of is transitive	$(\forall x)((\forall y)((\forall z)((\text{part-of}(x,y) \wedge \text{part-of}(y,z) \supset \text{part-of}(x,z)))))$	transitive(part-of)
R is transitive iff whenever R(x,y) and R(y,z) hold, R(x,z) holds too	not expressible (see however pseudo-second order)	$(\forall R)((\text{transitive}(R) \equiv (\forall x)((\forall y)((\forall z)((R(x,y) \wedge R(y,z) \supset R(x,z)))))))$

Higher order logic

Logic systems overview



Logical models - examples

Propositional Logic $A \wedge B \rightarrow C$

First Order Predicate Logic

$$\frac{P(a) \qquad (\forall x)(P(x) \rightarrow Q(x))}{Q(a)}$$

Description logics

Deals with concepts and relationships among these concepts

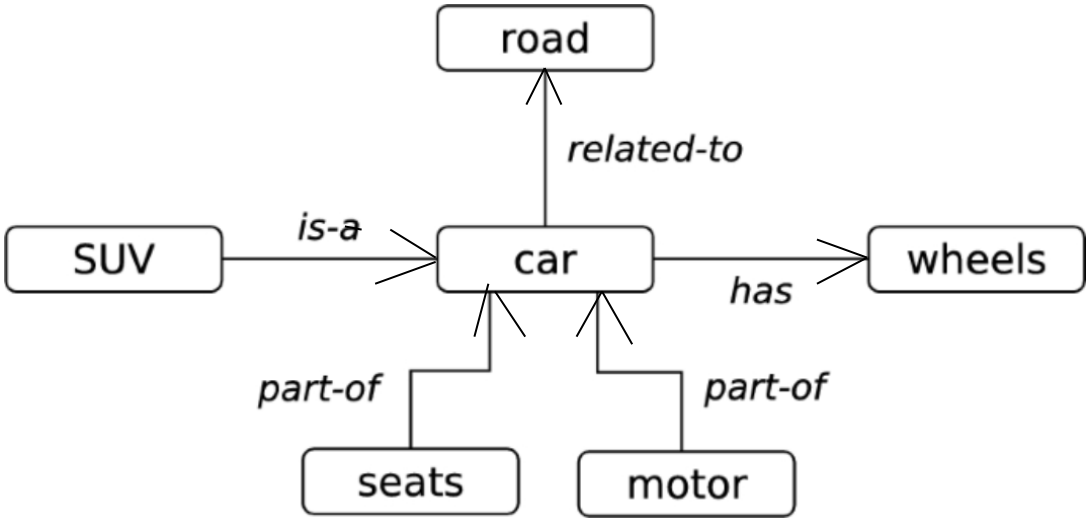
Woman ≡ Person ⊓ Female

Mother ≡ Woman ⊓ ∃hasChild.Person

MotherWithManyChildren ≡ Mother ⊓ ≥ 3 hasChild

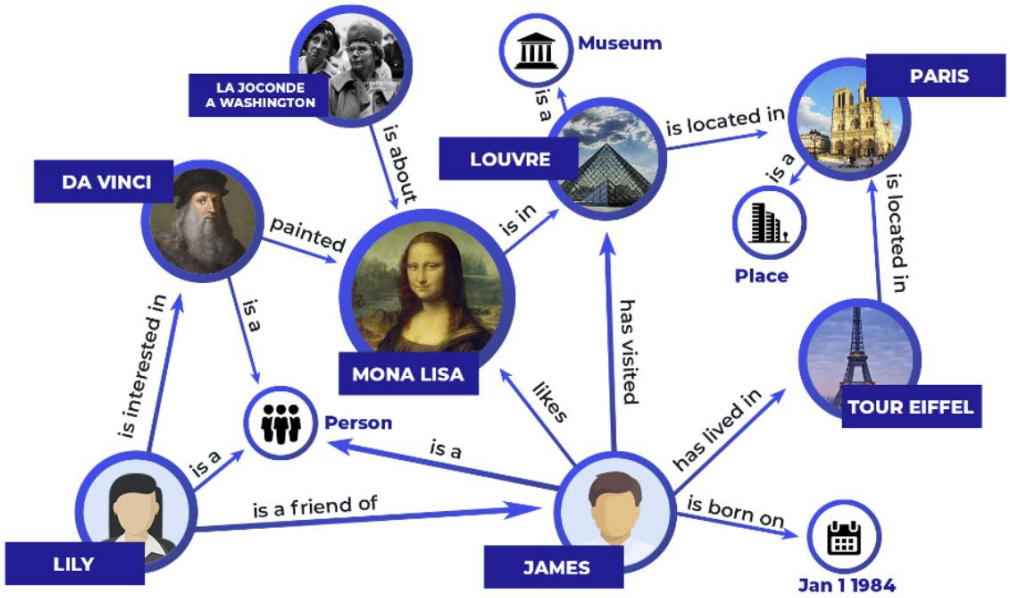
Semantic networks

- Knowledge represented as a **graph**
- **Nodes represent concepts or objects**, and **edges represent relationships** between them – **triples (SUV is-a car)**
- Initially used for representing NL meanings
- Currently, semantic networks are mostly used for **conceptual visualization** and **associative reasoning**



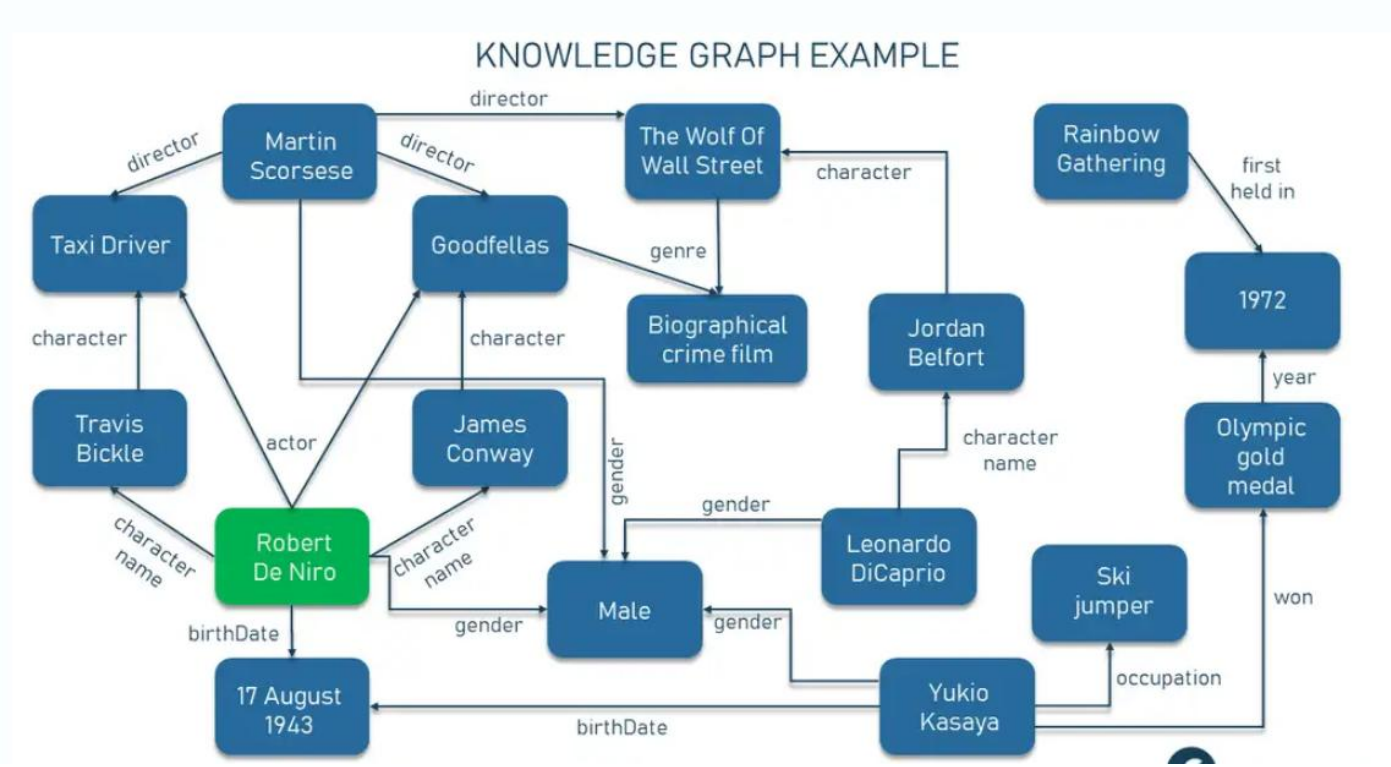
Knowledge Graphs

- The modern form of semantic networks, much larger
- A collection of **facts (triples)** about entities
- Can integrate data from multiple domains providing rich semantic understanding of information
- Built with technologies like **RDF, OWL, SPARQL**, often linked to the **Semantic Web**
- Used in applications such as search engines, recommendation systems, AI assistants



Knowledge graphs - examples

- ***Google*** uses its Knowledge Graph to enhance search results by providing users with facts and connections between entities
- ***Amazon*** and ***Netflix*** use knowledge graphs to recommend products based on user preferences and behaviors
- Assistants like ***Siri*** and ***Alexa*** use knowledge graphs to answer user queries based on the relationships between real-world entities



Knowledge graphs - evolution

- **2007 DBpedia** - DBpedia knowledge base was created as a result of the semantic processing of the Wikipedia article on infoboxes.
- **2012 Google Knowledge Graph -** Google Knowledge Graph - represents knowledge in graphs, coining the term “knowledge graph”
- **2012 Wikidata.** While DBpedia uses Wikipedia data to enrich the graph, the Wikidata graph uses an advanced knowledge modeling approach to describe entities and relationships.
- Wikidata currently contains over 95M entities (nodes) with millions of semantic relations between one another.

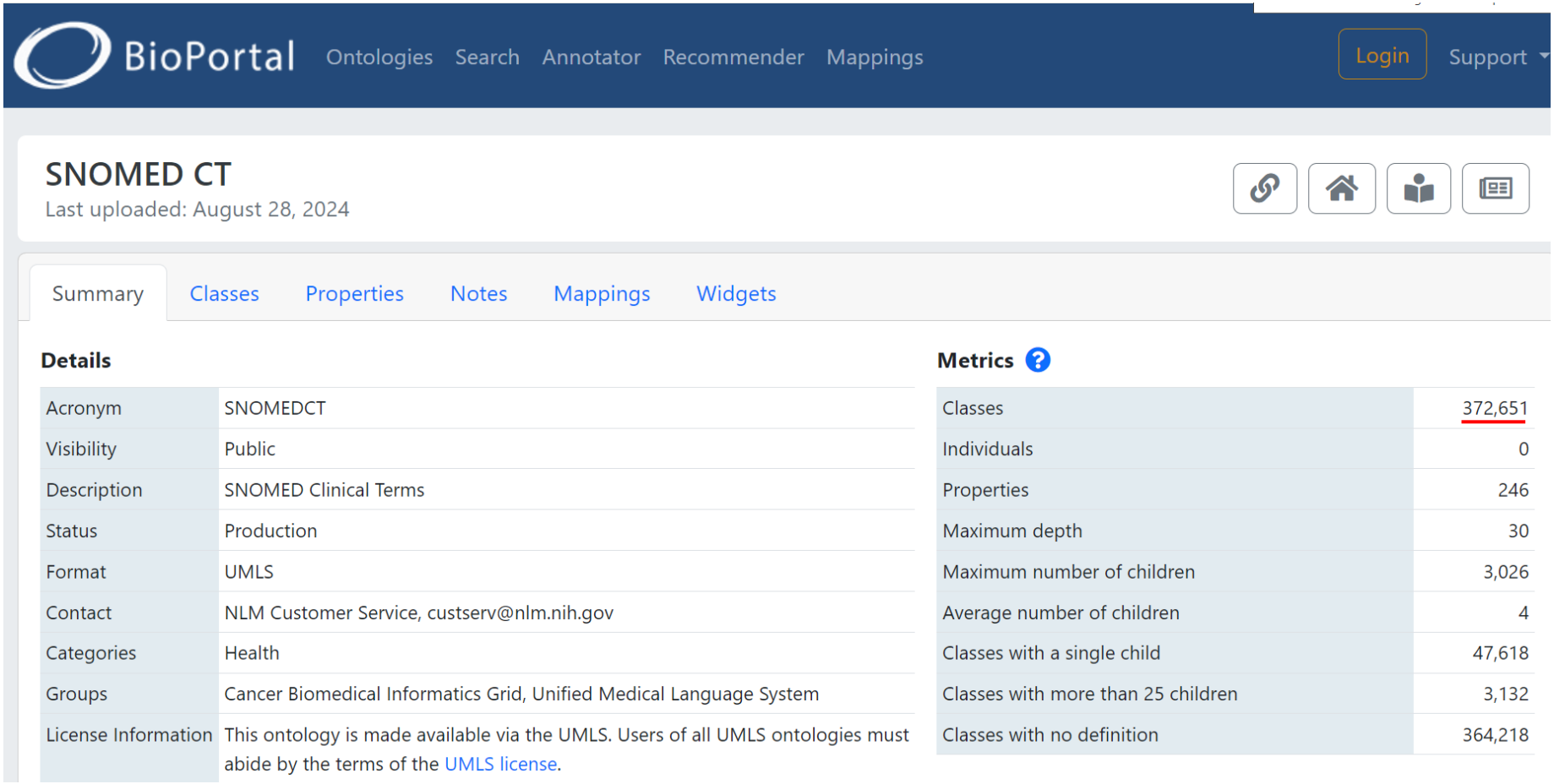
- Knowledge graphs available on the Internet form a Linked Open Data Cloud (LOD Cloud), semantically combining published graphs into one giant network.
- In 2007 this cloud consisted of 12 graphs
- In May 2020 it had grown to 1255 graphs in nine different domains.

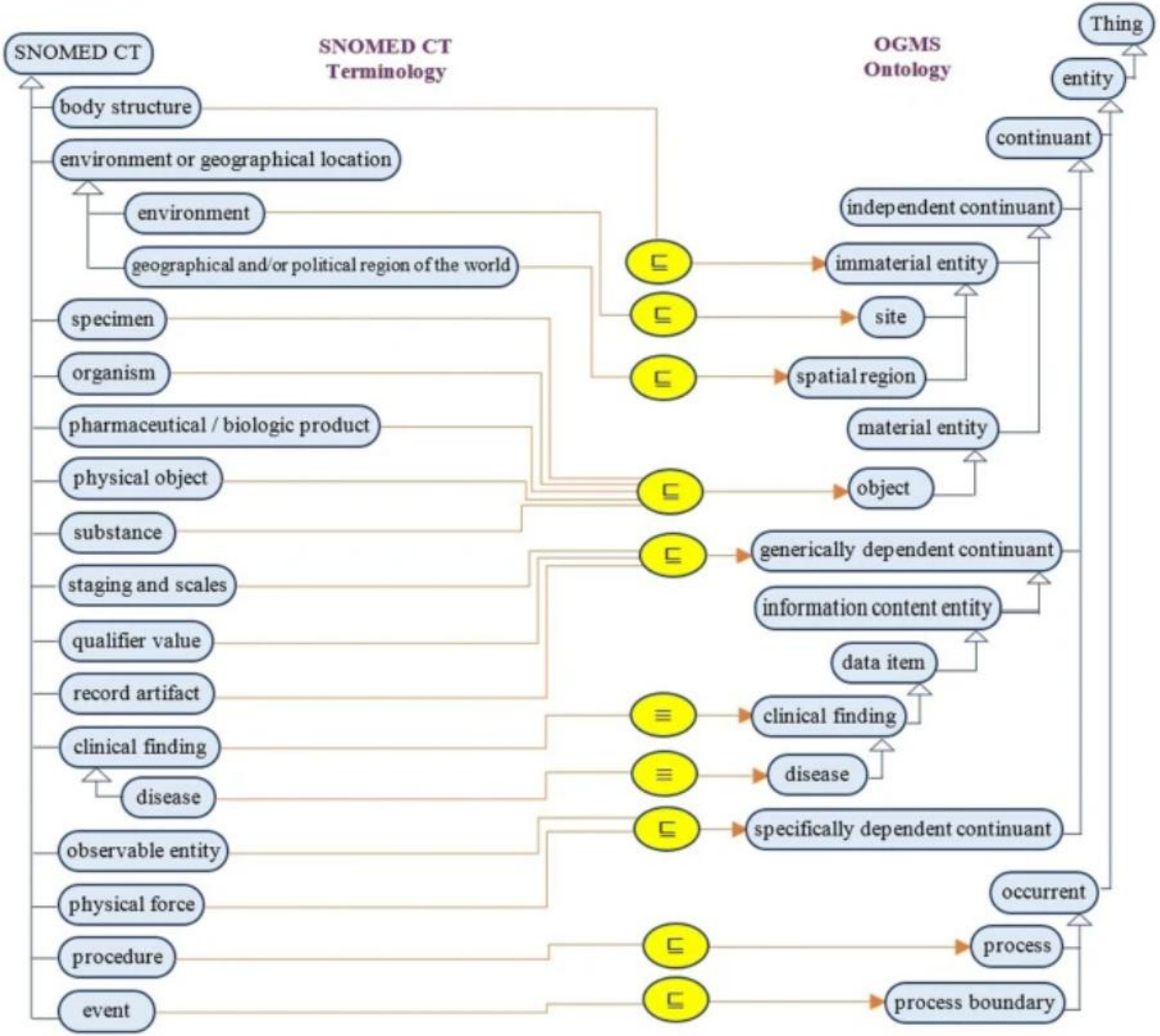
Ontologies

- In **philosophy**, **ontology** = the study of *what exists* and how things are categorized.
- In **computer science / AI**, **ontology** = a **formal specification of concepts and their relationships** in a given domain.
- It defines:
  - **Classes (concepts)** → e.g., *Animal, Dog, Cat*
  - **Properties (relationships/attributes)** → e.g., *hasColor, isA, eats*
  - **Constraints/axioms** → e.g., *All dogs are animals; No person can be both alive and dead.*
- Ontology description languages: **RDF Schema (RDFS), OWL (Web Ontology Language)**.
- Based on description logic
- Used for interoperability

Ontologies - examples

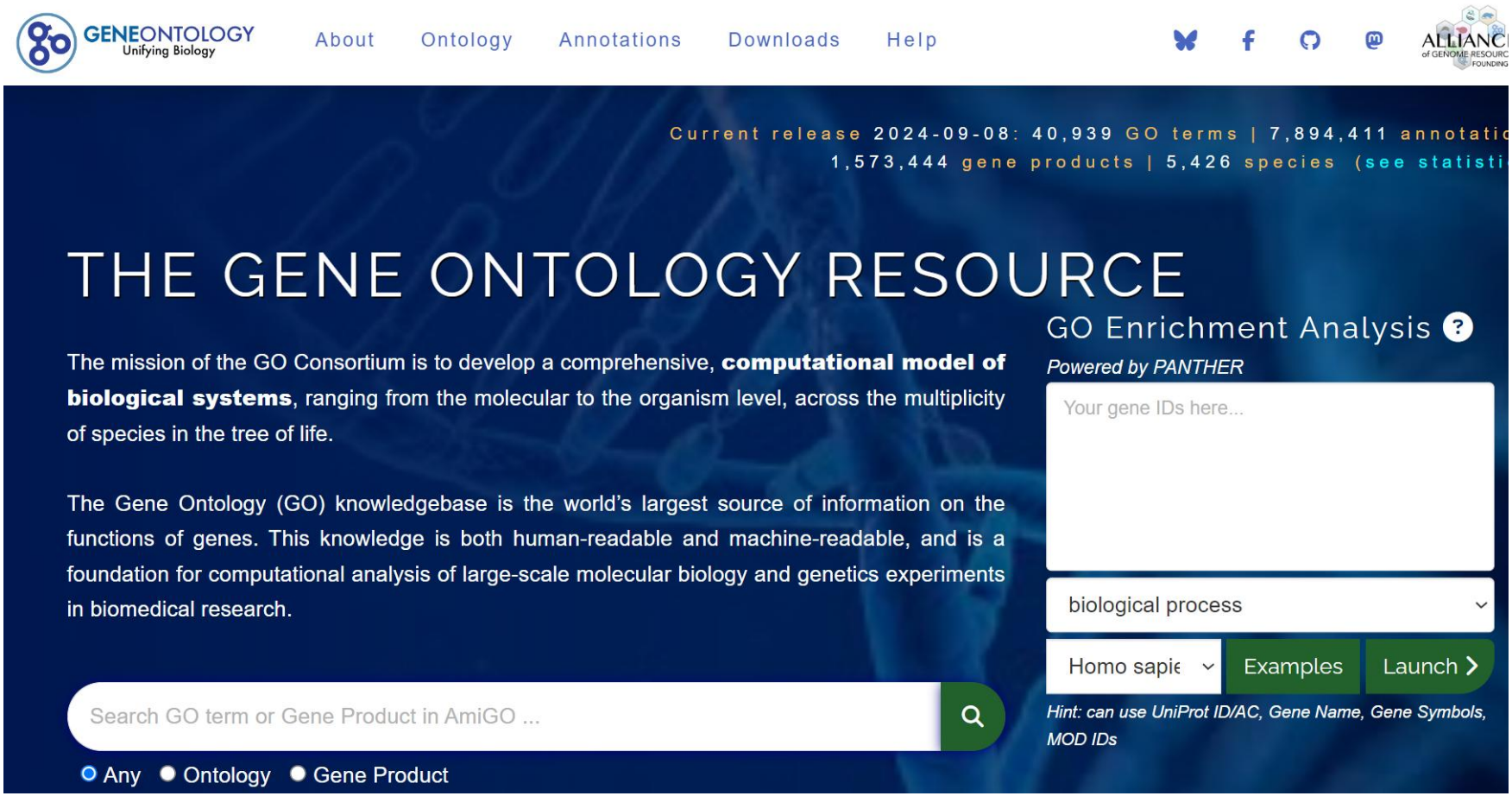
SNOMED CT Ontology

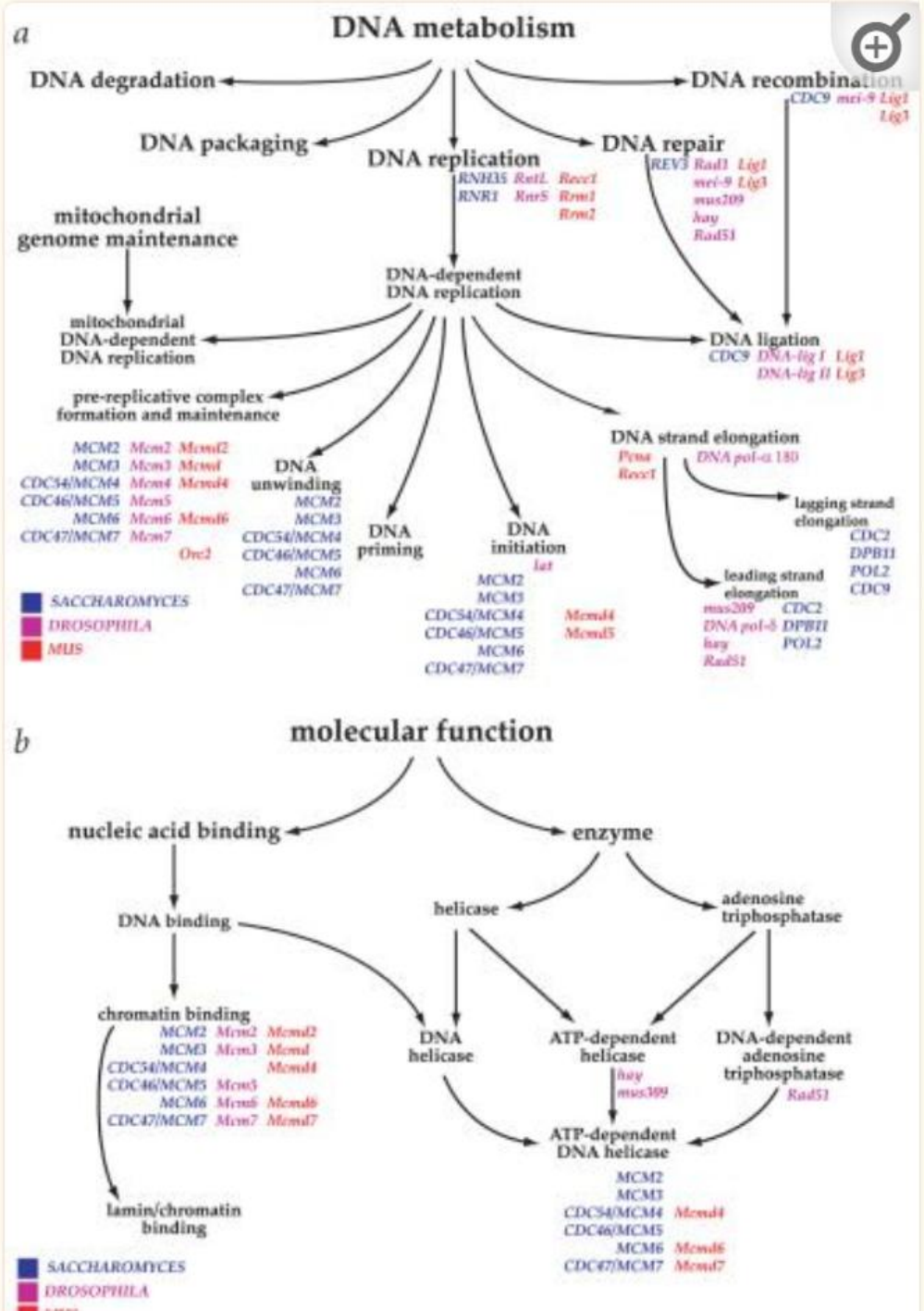



SNOMED CT
SNOMED CT
Terminology
OGMS
Ontology
Thing
entity
continuant
independent continuant
immaterial entity
site
spatial region
material entity
object
generically dependent continuant
information content entity
data item
clinical finding
disease
specifically dependent continuant
occurrent
process
process boundary
body structure
environment or geographical location
environment
geographical and/or political region of the world
specimen
organism
pharmaceutical / biologic product
physical object
substance
staging and scales
qualifier value
record artifact
clinical finding
disease
observable entity
physical force
procedure
event

Ontologies - examples

GeneOntology




a
DNA metabolism
DNA degradation
DNA packaging
DNA replication
DNA repair
DNA recombination
CDC9 mei-9 Lig1 Lig3
RNH35 RnrL RecC1 RNR1 Rnr5 Rnm1 Rnm2
REV3 Rad1 Lig1 mei-9 Lig3 mus209 hay Rad51
mitochondrial genome maintenance
DNA-dependent DNA replication
mitochondrial DNA-dependent DNA replication
DNA ligation
CDC9 DNA-lig I Lig1 DNA-lig II Lig3
pre-replicative complex formation and maintenance
MCM2 Mcm2 Mcmd2
MCM3 Mcm3 Mcmd
CDC54/MCM4 Mcm4 Mcmd4
CDC46/MCM5 Mcm5
MCM6 Mcm6 Mcmd6
CDC47/MCM7 Mcm7
Orc2
DNA unwinding
MCM2
MCM3
CDC54/MCM4
CDC46/MCM5
MCM6
CDC47/MCM7
DNA priming
DNA initiation
lat
MCM2
MCM3
CDC54/MCM4 Mcmd4
CDC46/MCM5 Mcmd5
MCM6
CDC47/MCM7
DNA strand elongation
Pcna
RecC1
DNA pol-α 180
lagging strand elongation
CDC2
DPB11
POL2
CDC9
leading strand elongation
mus209 CDC2
DNA pol-δ DPB11
hay POL2
Rad51
SACCHAROMYCES
DROSOPHILA
MUS
b
molecular function
nucleic acid binding
enzyme
DNA binding
helicase
adenosine triphosphatase
chromatin binding
MCM2 Mcm2 Mcmd2
MCM3 Mcm3 Mcmd
CDC54/MCM4 Mcmd4
CDC46/MCM5 Mcm5
MCM6 Mcm6 Mcmd6
CDC47/MCM7 Mcm7 Mcmd7
DNA helicase
ATP-dependent helicase
hay
mus309
DNA-dependent adenosine triphosphatase
Rad51
ATP-dependent DNA helicase
MCM2
MCM3
CDC54/MCM4 Mcmd4
CDC46/MCM5
MCM6 Mcmd6
CDC47/MCM7 Mcmd7
lamin/chromatin binding
SACCHAROMYCES
DROSOPHILA

Ontologies vs Knowledge graphs

Feature	Ontology	Knowledge Graph
Aim	Formal, logical structure for reasoning	Store and retrieve interconnected facts
Structure	Highly structured, formal	Looser
Formality	Based on description logic	May rely on ontologies but often less formal
Reasoning	Complex reasoning and inference	Basic reasoning and query
Data sources	Hand-crafted, domain specific	Integrates data from multiple sources
Flexibility	Requires logical consistency	More flexible
Integration	Focuses on a specific domain	Combines data from multiple domains
Applications	Sematic Web, expert systems	Search engines, recommendation systems, AI assistants

Rule based models

- We often have to deal with situations like

  if $Cond_1 \wedge \ldots \wedge Cond_n$

  then perform some actions / draw a conclusion

- Such a statement is called production rule / decision rule / if-then rule

- A **Rule Based System** (RBS) uses a set of such rules to infer conclusions or take actions based on given conditions

- They are based on a subset of first order logic

- These systems are often used in **expert systems** and **decision-making systems**

Rules - examples

Production Rules

Rule 1:
IF the **patient has a fever** AND **patient has a sore throat**
THEN the **diagnosis is possible flu**.

Rule 2:
IF the **patient has a rash** AND **patient has a fever**
THEN the **diagnosis is possible measles**.

Rule 3:
IF the **patient has a headache** AND **patient has nausea**
THEN the **diagnosis is possible migraine**.

Rule 4:
IF the **patient has a cough** AND **patient has shortness of breath**
THEN the **diagnosis is possible bronchitis**.

Rule 5:
IF the **patient has a runny nose** AND **patient has sneezing**
THEN the **diagnosis is possible common cold**.

- This is the NL form of rules. They are coded using a rule language quite close to the NL form

Rule based systems / languages

Classic but very influential

- **OPS5** - Official Production System
- **MYCIN**
- **CLIPS** - **C Language Integrated Production System** (still maintained)

Current

Semantic Web

- **SWRL - Semantic Web Rule Language** (a W3C Member Submission, never a standard), RIF, SHACL Rules, N3
- **RuleML** – **Rule Markup Language** (Rule Markup Initiative)

Logic programming / Datalog: Prolog, ASP, Soufflé, RDFox

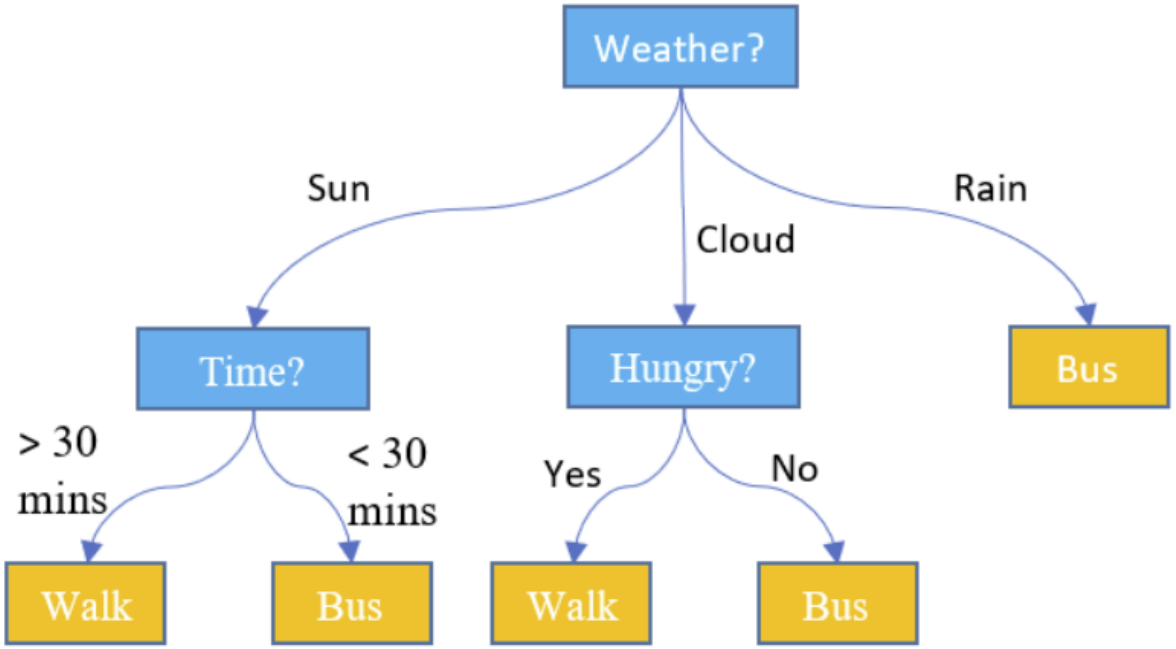
Business rules

- **Drools** - **Business Rules Management System**
- **Drools** is a **RBS** and a core component of the **JBoss Rules** framework
- **JBoss Enterprise BRMS (Business Rules Management System)** is a **Red Hat product** built around Drools. It provides a business rules management system that combines the Drools rules engine with additional features for enterprise-level rule management

Classification models

Decision trees, Random forests

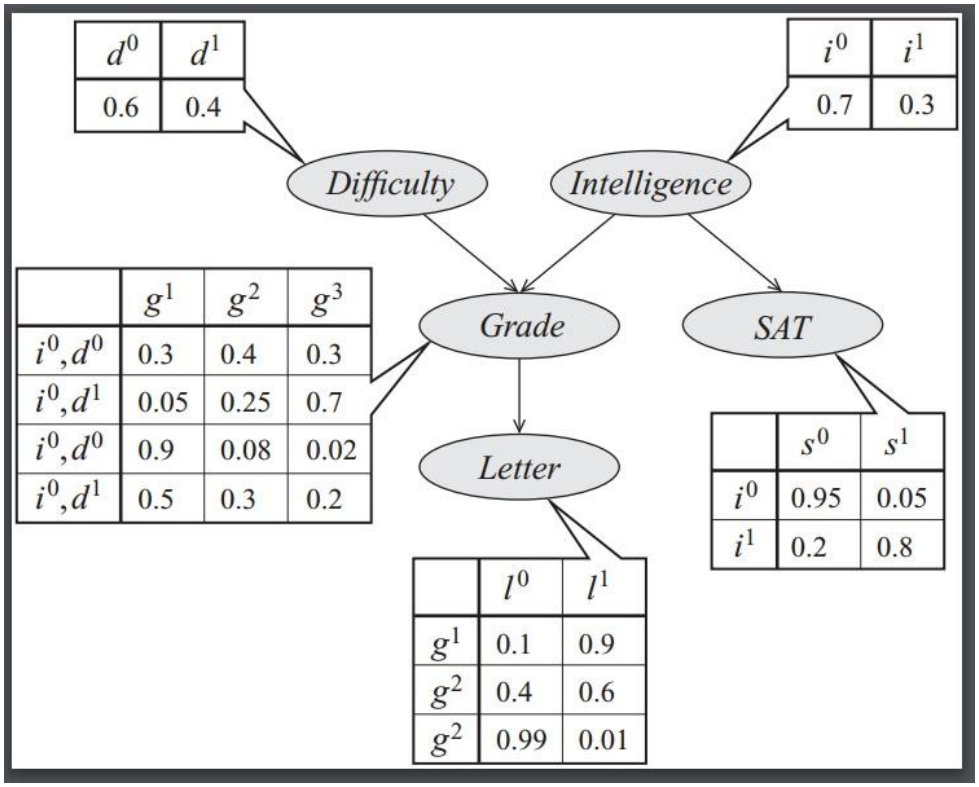
- **Decision trees** - a **general tree** used for **classification tasks**, where the goal is to predict a class from a given input.
- Categorial outcomes



Probabilistic models

Bayesian networks

- A **Bayesian network** (**Bayes network**, **Bayesian belief network**, or **probabilistic graphical model**) is a statistical model that represents a set of variables and their probabilistic dependencies using a directed acyclic graph (DAG).

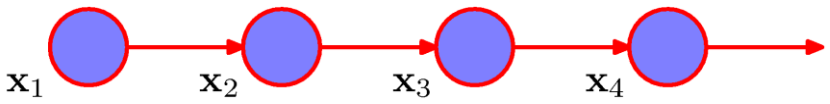


Markov chains

- A **Markov chain** is a stochastic model describing a sequence of possible events in which the probability of each event depends only on the state attained in the previous event(s).

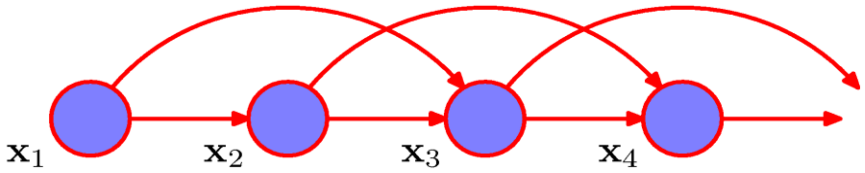


Andrey Markov
1856-1922



$$p(\mathbf{X}) = \prod_{i=1}^{n} p(X_n|X_{n-1})$$

1st order

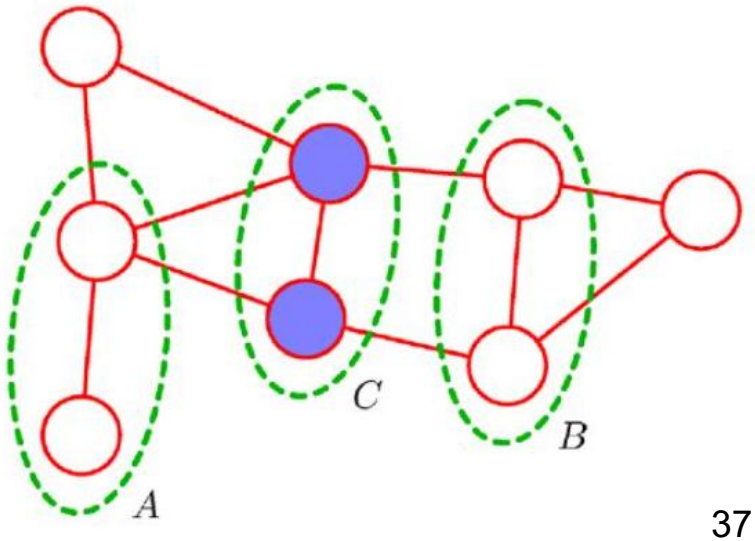


$$p(\mathbf{X}) = \prod_{i=1}^{n} p(X_n|X_{n-1}, X_{n-2})$$

2nd order

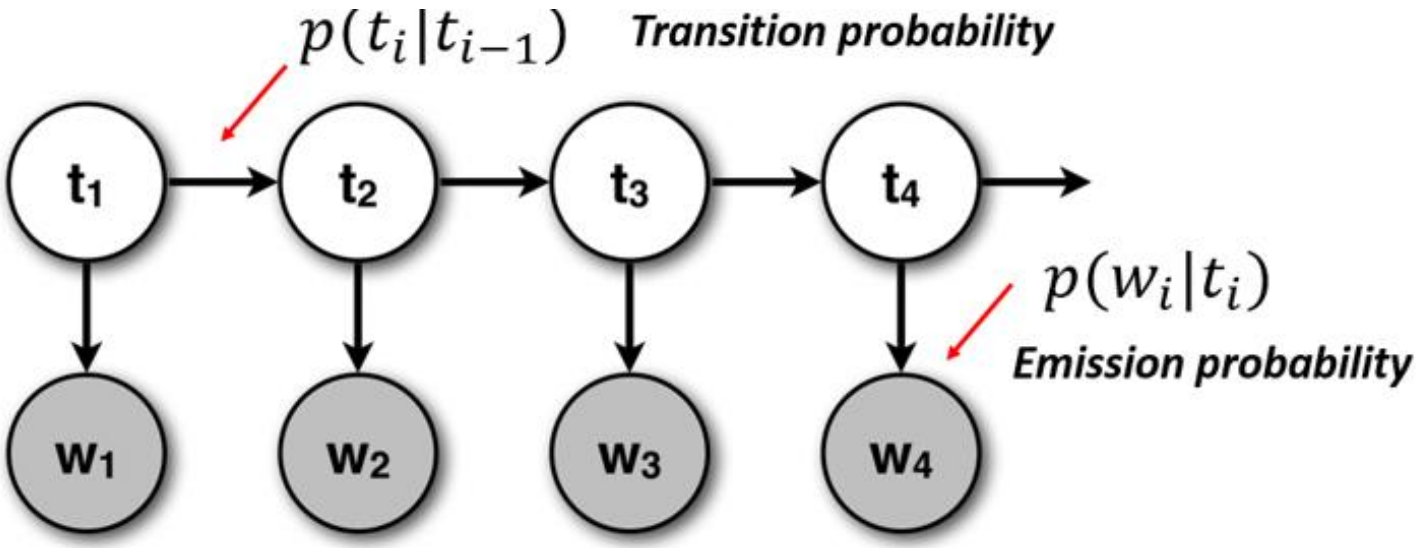
Markov networks

- **Markov network** or **Markov Random Field** - a type of **probabilistic graphical model** that represents the joint distribution of a set of random variables using an **undirected graph**.
- Unlike Bayesian networks, which use directed edges to represent dependencies, Markov networks use **undirected edges** to model relationships and encode conditional independence between variables.
- Markov networks may be cyclic



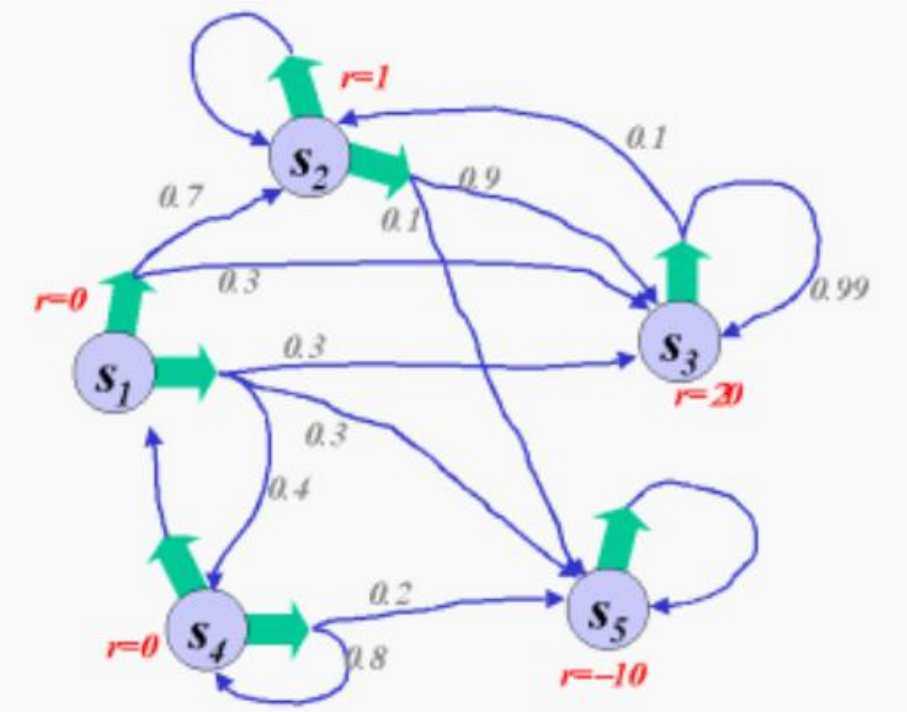
Hidden Markov Models

- A **Hidden Markov Model** (HMM) is a statistical model used to represent systems that follow a **Markov chain** with **hidden states**.
- It is widely used in areas involving time-series data or sequential data.



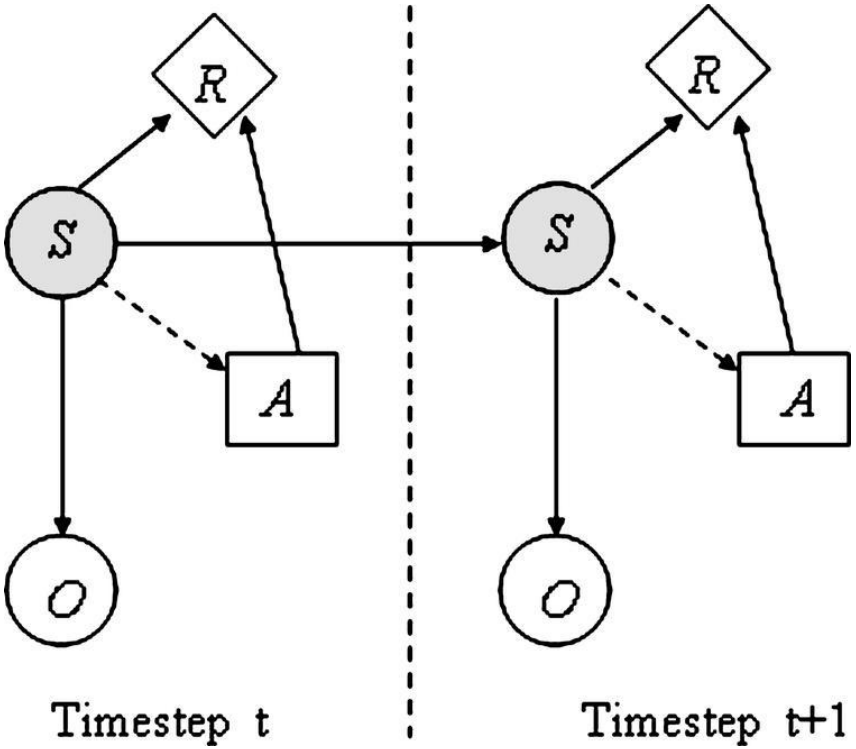
Markov Decision Processes

- A **Markov Decision Process** (MDP) is a Markov chain in which state transitions depend on the current **state** and on the **action** applied to the system.
- A Markov decision process is used to compute a policy of actions that will maximize some utility with respect to **long-term rewards**.
- Used as a basis for **Reinforcement Learning** (and other)



Paryially Observable Markov Decision Processes

- A **Partially Observable Markov Decision Process** (POMDP) is a Markov decision process in which the states of the system are **not all observable** (the system is partially observable)
- POMDPs are known to be NP complete, but approximation techniques make them useful for different applications, such as planning, robot control, and others
- Used very much in robotics

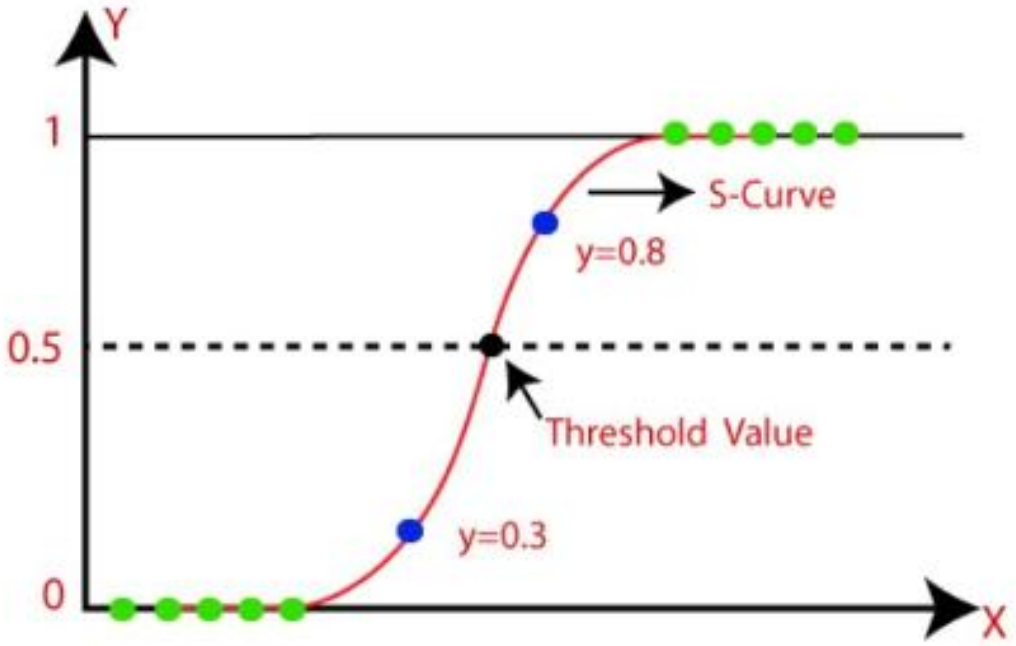


Non-symbolic models

Regression models

Logistic regression

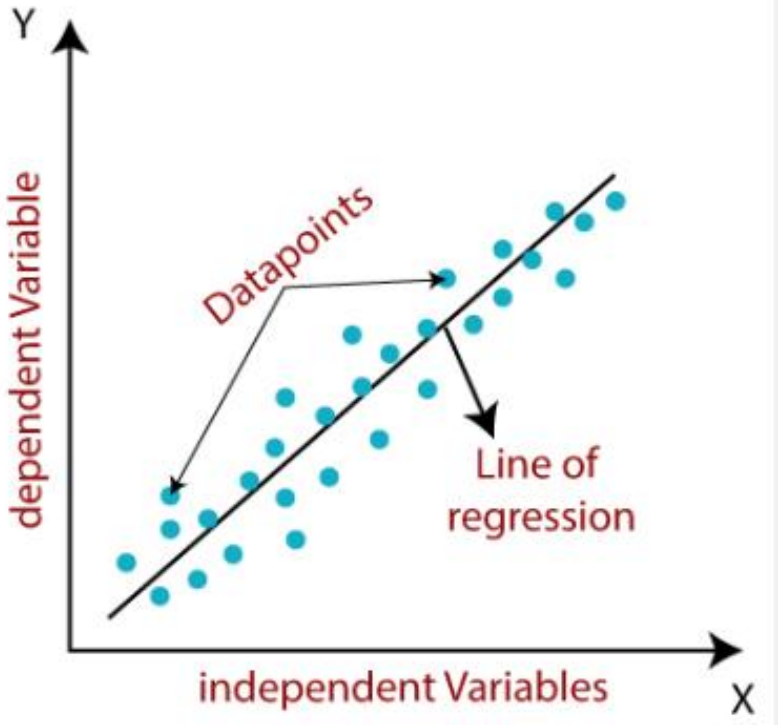
- A statistical model used for **binary classification** tasks, where the goal is to predict one of two possible outcomes (e.g., yes/no, true/false, 0/1).
- Estimate the probability that a given input belongs to a particular class.
- Logistic regression is used for **classification problems**, not regression problems.



Regression models

Linear regression

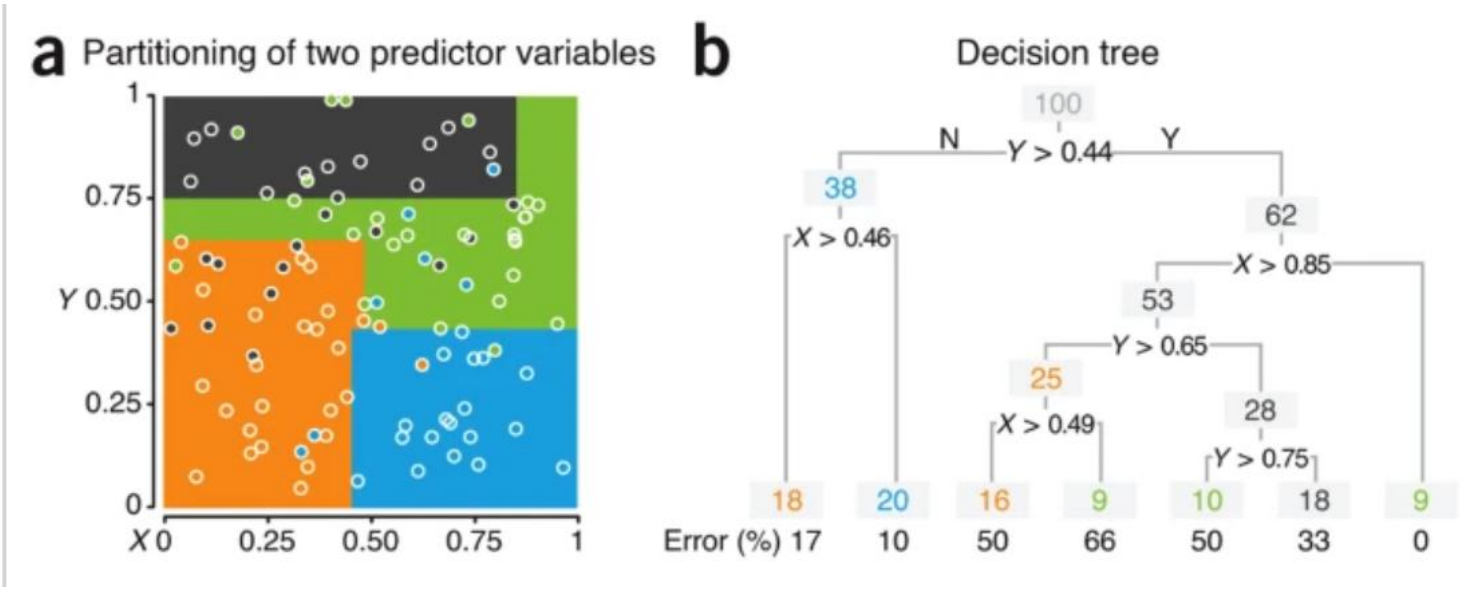
- **Linear regression** is a statistical method used to model the relationship between a **dependent variable** (also called the target or outcome) and one or more **independent variables** (also called predictors or features).
- Predict the value of the dependent variable based on the values of the independent variables by fitting a **linear relationship** between them.



Regression models

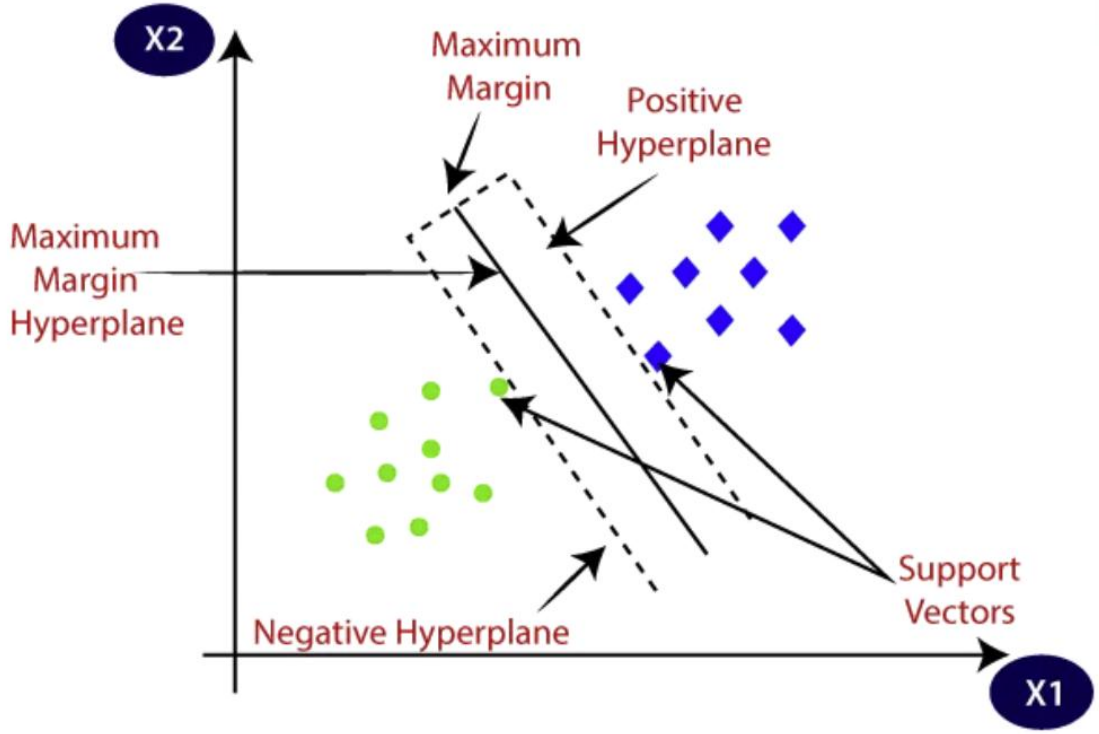
Regression trees

- **Regression trees** are a type of **decision tree** used for **regression tasks**, where the goal is to predict a continuous (numerical) target variable.
- In contrast to classification trees (which are used for categorical outcomes), regression trees aim to model and predict real-valued outputs, such as predicting a house price or a person's income based on features.



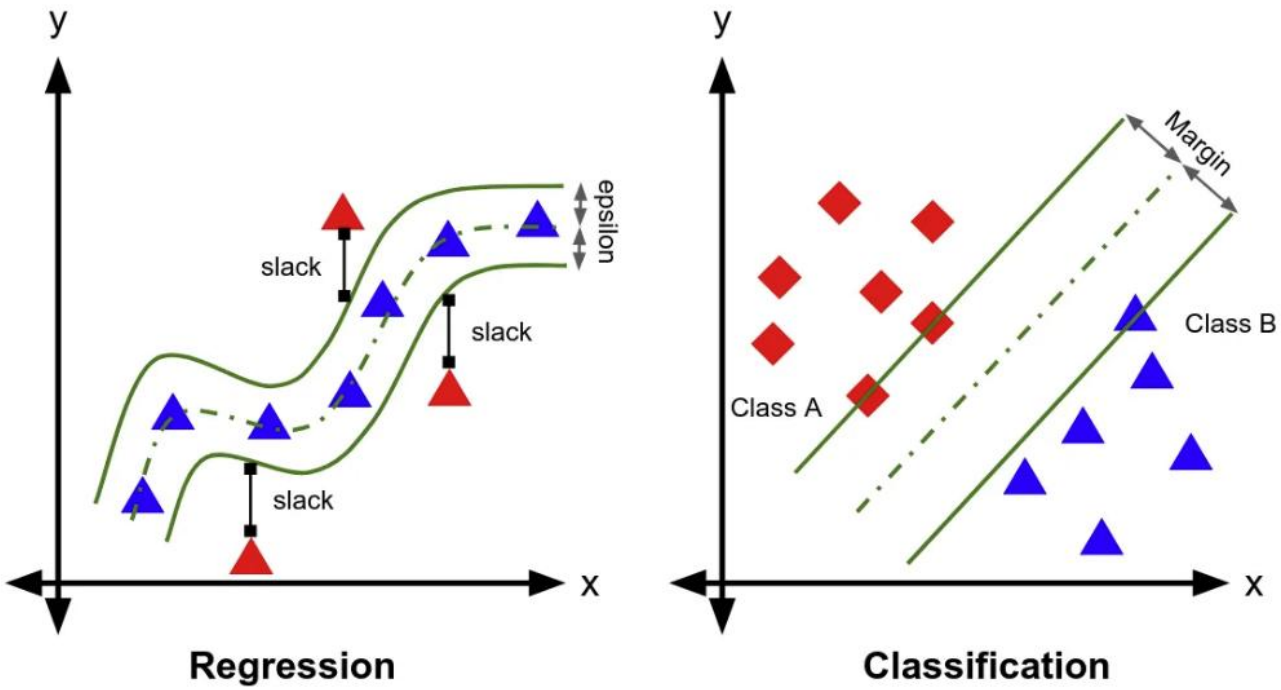
Support Vector Machines

- **SVM for classification** represents data as points in a high-dimensional vector space.
- It finds a **hyperplane** (mathematical decision boundary) that separates classes.
- Knowledge = stored in the **support vectors + weights** (numerical form)



Support Vector Regression

- **SVR is an extension of SVM for regression tasks**.
- It learns a function that approximates the relationship between input and output, in a numerical vector space.
- Knowledge = embedded in the learned function and support vectors.
- Goal = predict continuous values by finding the best-fitting function that predicts the target variable based on input features.

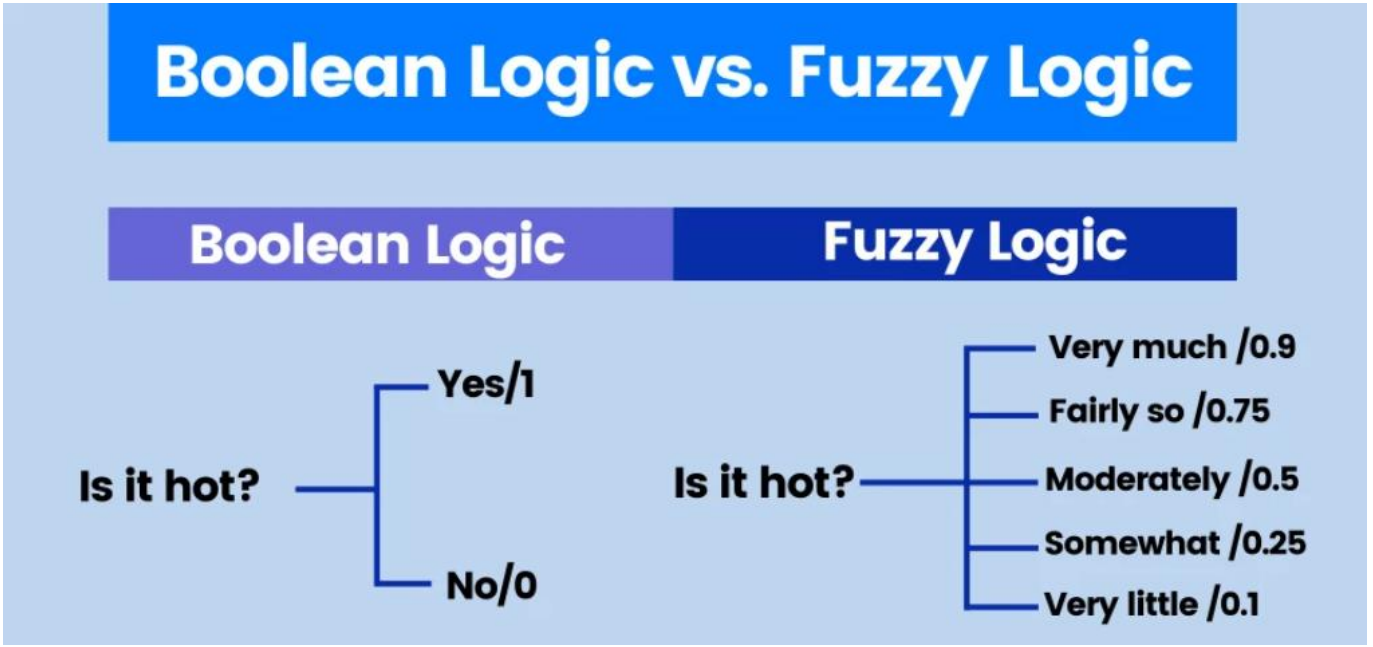


Fuzzy logic

- **Fuzzy logic** is a form of logic used to handle the concept of **partial truth:**

- the truth value of variables may be **any real number between 0** (completely false) and **1** (completely true)

- In fuzzy logic, an element can **partially** belong to a set.
- Fuzzy sets are defined with a **membership function** that assigns a degree of membership, typically between 0 and 1, to each element.



Fuzzy logic

Example

- In classical logic, a temperature might be either "hot" or "cold."
- In fuzzy logic, a temperature of 75°F might be considered "***0.7 hot***" and "***0.3 cold***," meaning it partially belongs to both the "hot" and "cold" sets.

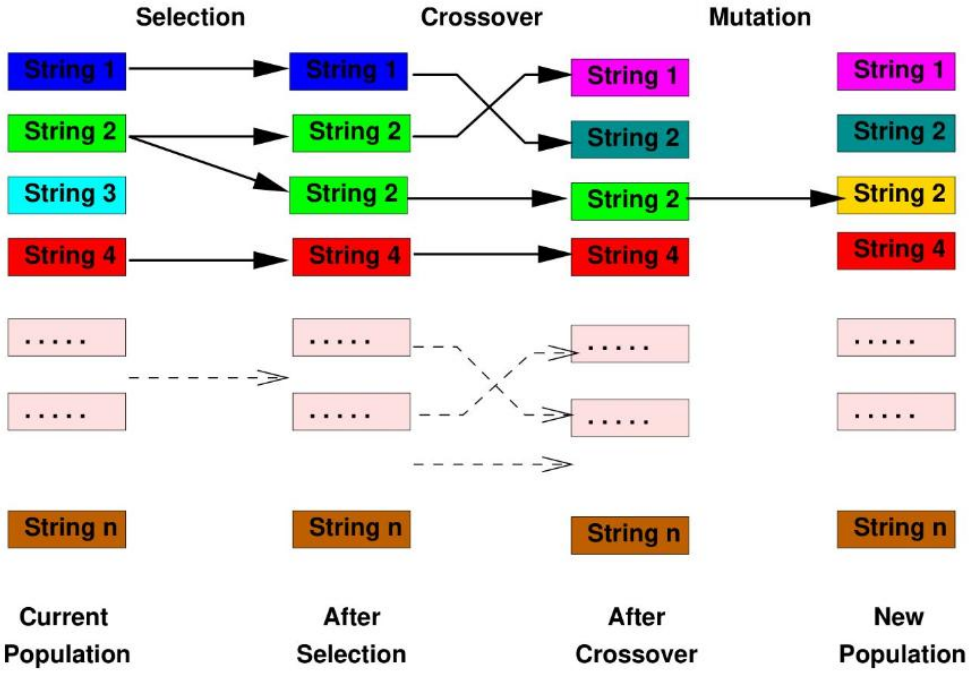
- Fuzzy logic systems are often governed by **fuzzy if-then rules**.
- **Example of a fuzzy rule**

IF the temperature is high AND the humidity is low, THEN turn the fan speed to high.

- The terms "high" and "low" are fuzzy variables with degrees of truth rather than precise numerical values.

Genetic models

- **Genetic model** - a model inspired by the principles of **biological evolution**.
- Implemented through **genetic algorithms (GAs)** or **genetic programming (GP)**, which simulate the process of natural selection and evolution to solve **optimization and search problems**.
- Used in: optimization, machine learning, robotics, and bioinformatics.
- Require careful parameter tuning to perform well in practice.



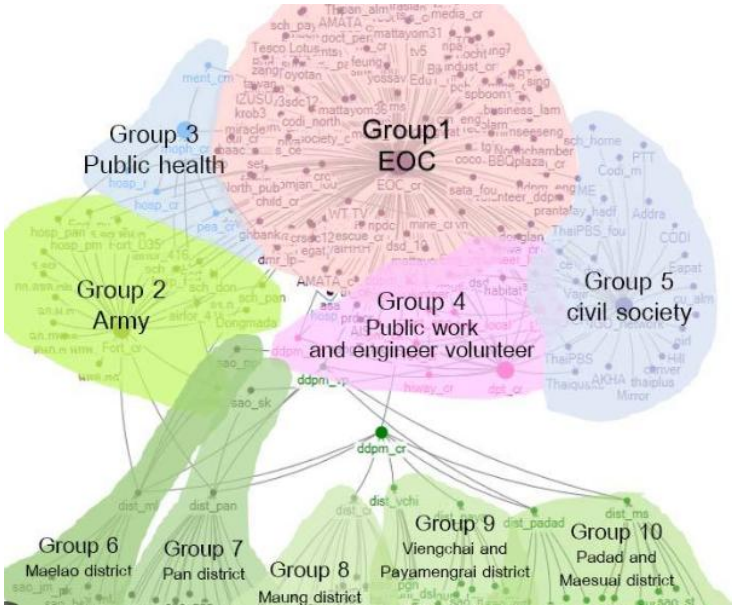
Clustering models

K-means clustering, hierarchical K-means

- K-means clustering aims to partition a dataset into **K clusters**.
- Each cluster contains data points that are more similar to each other than to those in other clusters.
- The similarity is measured by the distance (typically **Euclidean distance**) between points.



Customer clustering

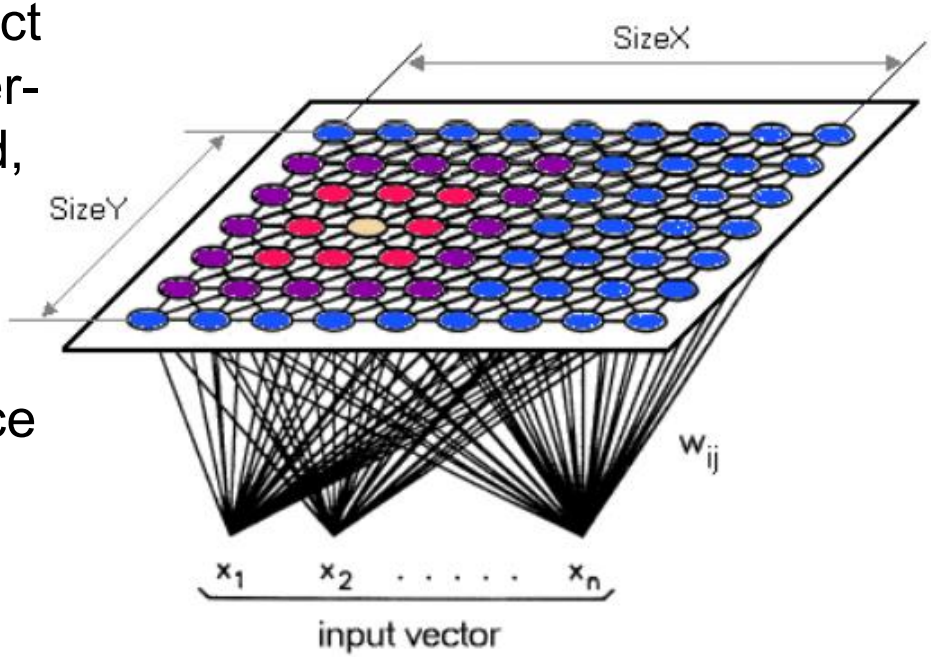


Document clustering

Clustering models

Self organizing neural networks

- **Self-Organizing Map** (SOM) – a neural network that learns to project high-dimensional data onto a lower-dimensional grid, usually a 2D grid, while maintaining the **topological relationships in the data**.
- Data points that are close to each other in the high-dimensional space will also be close to each other on the grid.
- Typically used for tasks like clustering, visualization, and dimensionality reduction



Distributed representations

Neural networks

- KR is **represented (implicitly) in weights and activations** of the layers
- **Output layer –** solution

Deep Neural Networks

- **Embeddings** - words, images, or concepts are mapped to dense vectors in a high-dimensional space
- **Hidden layers -** Capture abstract features and relationships learned from data

See NN course

Distributed representations

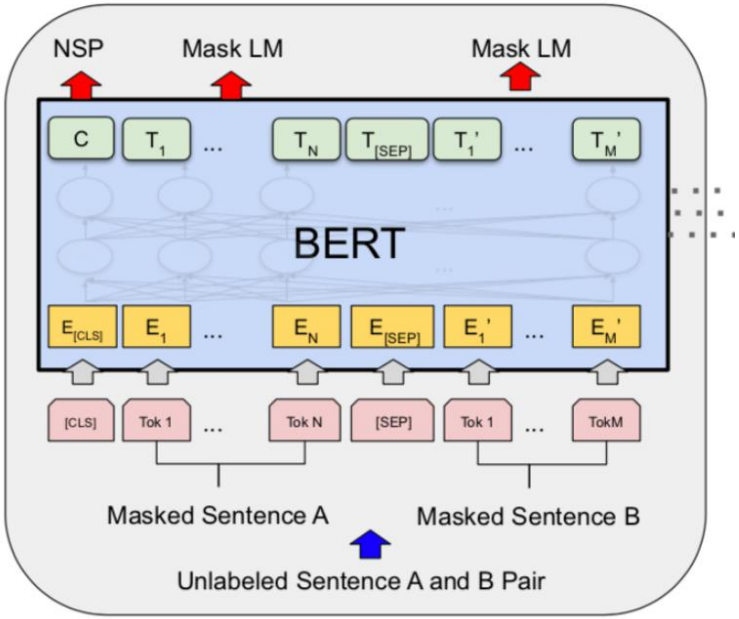
Vector-based representation (**embeddings**) of data such as words, images, nodes

- Vector-based representations that capture **semantic relationships**
- A way of **encoding information in a high-dimensional space**, where the meaning of the representation is spread across different dimensions.
- Different dimensions capture different **latent features** or aspects of meaning
- **Latent** = *hidden, not directly observable*.
- In ML, latent features are **not given explicitly in the input data**, but are **inferred, discovered, or learned** by the model

Distributed representations

Words/Text embeddings

- Concepts (such as words, phrases, or entities) are represented as vectors in a continuous vector space
- Distributed representations encode **semantic similarity**
- Word2Vec
- GloVe (Global Vectors for Word Representation)
- BERT (Bidirectional Encoder Representations from Transformers)



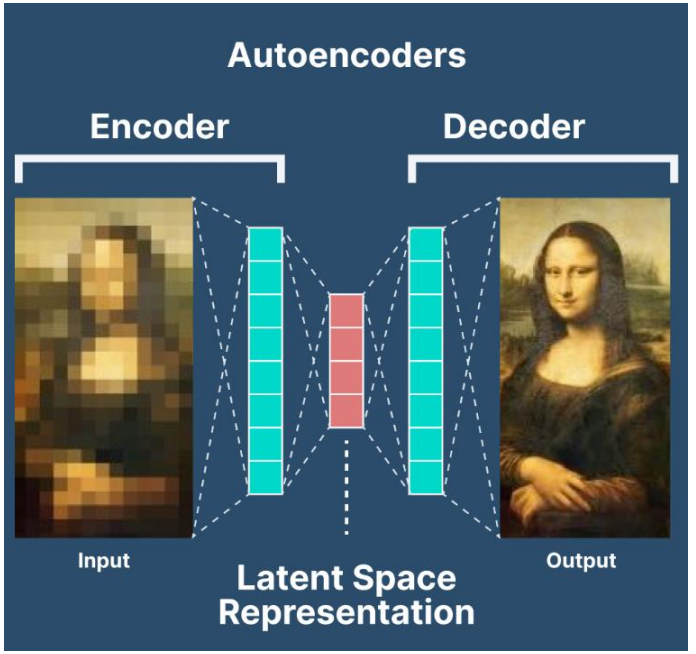
Distributed representations

General: image, audio, text, time-series data

Autoencoders

- NN that learns a compressed, efficient representation of input data - **encoding** - and then reconstruct the original data from this compressed version
- Used in tasks like data compression, denoising, dimensionality reduction, and feature extraction

VAE - generative modeling,
generate new data points similar to the input data (e.g., generating new images).



Distributed representations

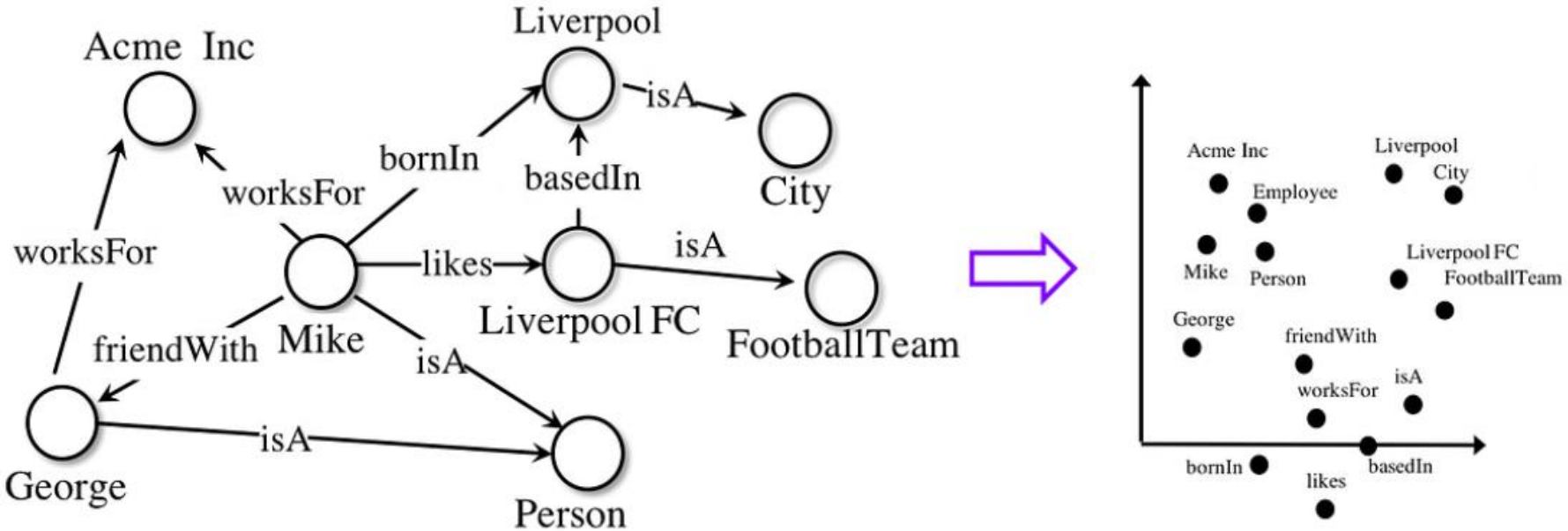
- **Word based distributed representations, such as BERT,** are designed for **text data**
- The learn distributed representations that capture linguistic and contextual relationships between words or tokens.
- Are **high-dimensional** and **context-sensitive**.
- Goal: natural language processing

- **Autoencoders and VAEs** are general-purpose models that can be applied to various types of data: **images**, **audio**, **text**, and **time-series data**.
- The distributed representation learned is often lower-dimensional than the input data, and its primary purpose is **compression** or **generation**.
- Goal: find a compact and efficient encoding that can be used for tasks like **feature extraction**.

Distributed representations

Knowledge graph embeddings

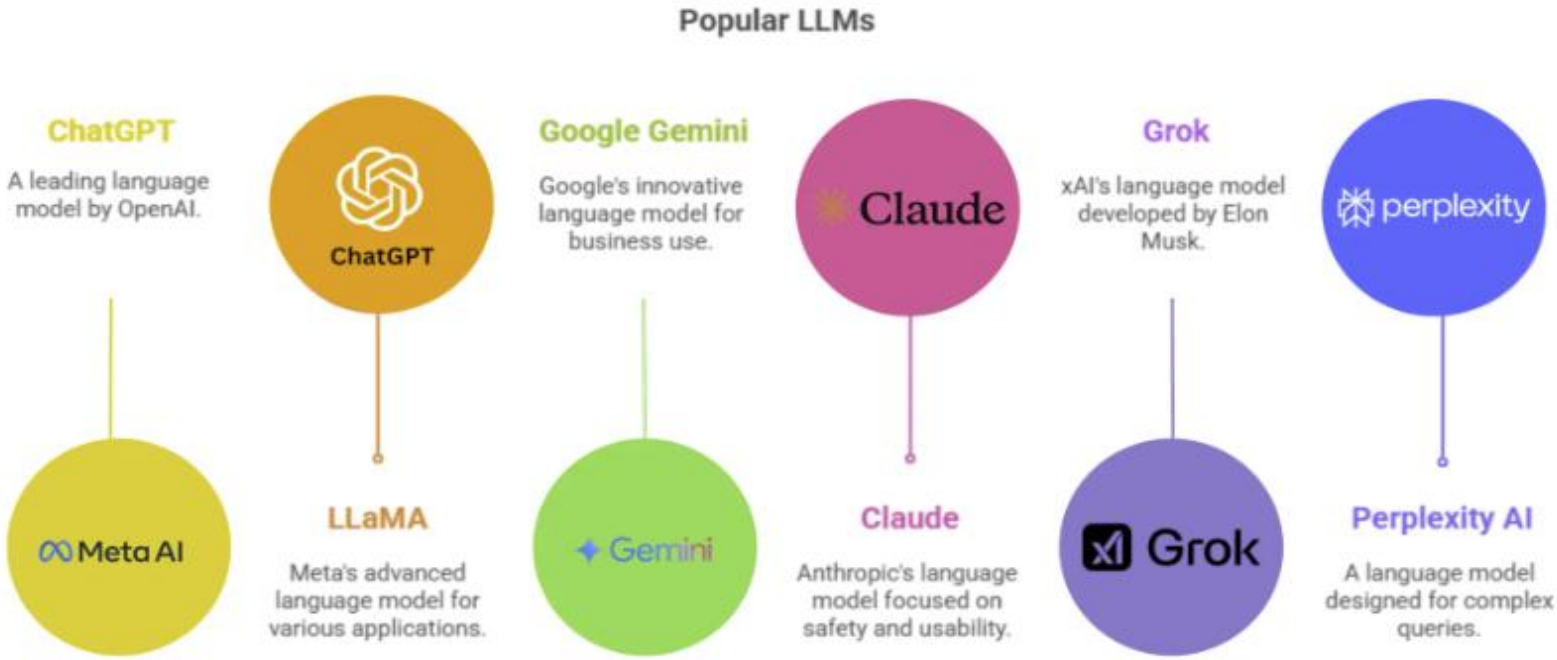
- Knowledge graphs are **discrete** and often **large-scale** → difficult to use them directly with ML models that expect **fixed-size input vectors**.
- Knowledge graph embeddings → map the entities and relationships in the graph to continuous vector spaces.



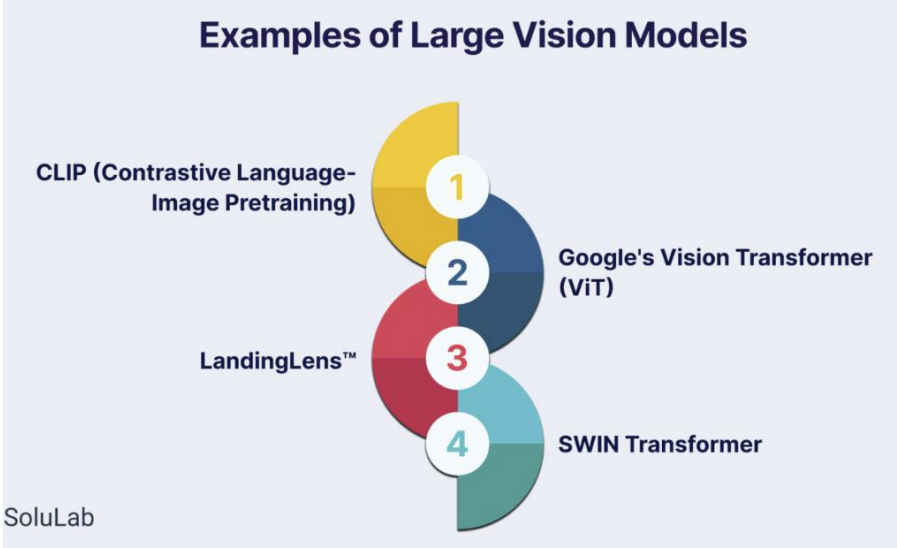
Distributed representations

Large Language Models (LLMs)

- KR → **Embeddings** (vectors, similar words are clustered), **Hidden layers** (abstract features, distributed), **Weights** (capture statistical co-occurrences across massive text corpora)
- Implicit reasoning via **learned language patterns** and **representation alignment** in a high-dimensional space



Distributed representations



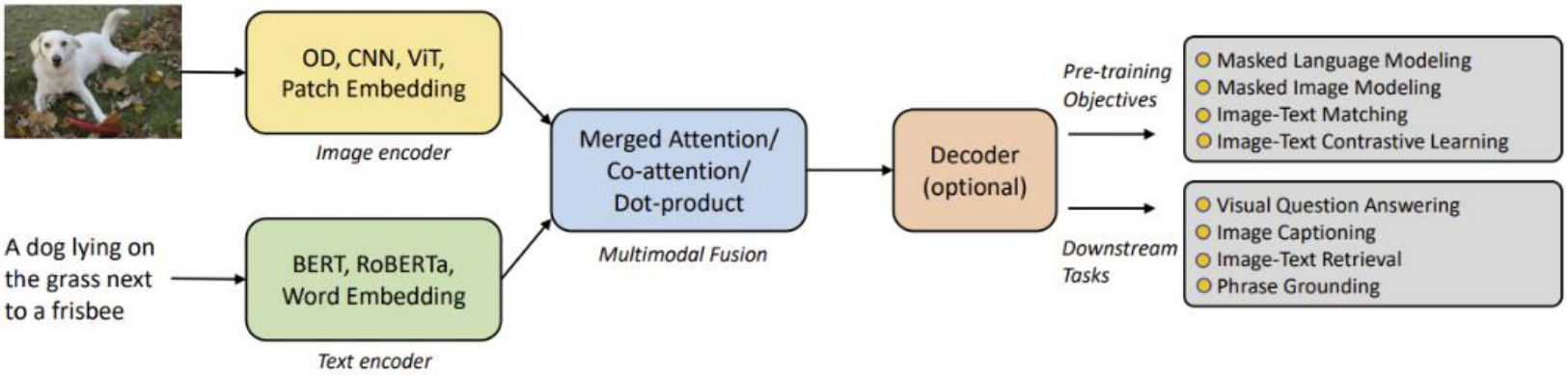
Large Vision Models (LVMs)

- KR → **Embeddings** (similar images are clustered), **Hidden layers** (**distributed features** in layers - early layers: capture visual primitives, middle layers: capture abstract patterns, later layers: capture semantic object-level concepts), **Weights** - shape how input pixels → embeddings
- Implicit reasoning via learned **visual pattern recognition** and **representation alignment** in a high-dimensional space

Distributed representations

Vision Language Models (VLMs)

- Involves grounding language in perception, e.g., “The boy is holding an umbrella, so it might be raining.”
- Blend **visual inference** with **language-based reasoning**.
- **Visual embeddigs** → capture **latent features** like edges, textures, objects, or even abstract concepts
- **Text embeddings** → capture semantics of text, just like in LLMs.
- **Cross-modal Representation** - a shared latent space aligns visual and textual features
- A **fusion mechanism** (projection layers, cross-attention, alignment loss, contrastive learning, etc.) → aligns both modalities



Distributed representations

VLMs (Vision Language Models)

OpenAI

- GPT-4V (Vision) → ChatGPT with vision (can analyze images + text).
- CLIP → Connects images with text (used for retrieval, classification).
- DALL·E 3 (partially VLM, though mainly generative from text).

Google DeepMind

- Gemini (multimodal) → successor of PaLM-E, integrates vision + language + reasoning.
- Flamingo → visual question answering and captioning model.
- PaLI / PaLI-X → Vision-language for many languages + tasks.

Distributed representations

VLMs (Vision Language Models)

Meta (Facebook AI)

- LLaVA (Large Language and Vision Assistant) → Combines LLaMA + visual encoder.
- ImageBind → Aligns images, text, audio, depth, thermal, IMU signals into a shared space.
- Kosmos-1 → Multimodal model that can understand images + text prompts.

Microsoft

- Kosmos-2 (with grounding: connecting language to objects in images).
- Florence-2 → Vision-language foundation model for image captioning, classification, grounding.

Distributed representations

Future

- **Hybrid systems**

Combine **statistical LLM-style reasoning** with **formal symbolic reasoning** to get the best of both worlds.

